Boundary circulation for an inertial projection scheme

Abstract

We study the fixed-step recurrence $x_{n+1} = P_C(2x_n - x_{n-1})$ on a compact planar convex body with C^5 boundary and strictly positive curvature. Every sufficiently close distinct boundary pair generates a trajectory that preserves its orientation and circulates indefinitely. Its increments vanish, while its position limit set is the entire boundary. If ℓ is the perimeter and I is the boundary integral of squared curvature, then the squared chord increment satisfies $n\|x_n - x_{n-1}\|^2 \to \ell/(2I)$. Both accumulated arclength and chord length grow as $\sqrt{2\ell/I}\sqrt{n}$, and the iteration-sampled positions converge in distribution to normalized arclength. The proof derives a uniform cubic phase loss from the actual metric projection and establishes averaging by summation by parts. The recurrence and its artificial damping belong to earlier impact and position-projection schemes; the result concerns their fixed-step long-time boundary dynamics.

Keywords: discrete dynamical system; inertial projection; boundary circulation; numerical dissipation; occupation measure

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1 The discrete system and its boundary motion

Let $C \subset \mathbb{R}^2$ be a compact convex body with nonempty interior, and let P_C denote Euclidean metric projection onto the filled body. We consider

$$x_{n+1} = P_C(2x_n - x_{n-1}), \qquad n \geq 0. \tag{1}$$

The state consists of two consecutive positions. A constant state is fixed, since $P_C x = x$ for $x \in C$. The question here is how a small nonzero boundary displacement evolves: whether it disappears after finite travel, and which part of the boundary the resulting trajectory visits.

Assume that ∂C is C^5 and has strictly positive curvature. Write

$$r : \mathbb{R} \longrightarrow \partial C, \qquad \|r'(s)\| = 1, \qquad r(s+\ell) = r(s), \qquad I = \int_0^\ell \kappa(s)^2 \, \mathrm{d}s, \tag{2}$$

where r is one-to-one modulo the perimeter ℓ, and $\kappa(s) = \|r''(s)\| > 0$. Either orientation may be chosen. For a bounded sequence, its ω-limit set denotes the set of limits of its subsequences with indices tending to infinity.

Theorem 1. *There is $a_* = a_*(C) \in (0, \ell/8)$ such that the following assertions hold for every $s_0 \in \mathbb{R}$, either orientation of r, and $0 < a_0 < a_*$. Initialize* (1) *by*

$$x_{-1} = r(s_0 - a_0), \qquad x_0 = r(s_0). \tag{3}$$

Every extrapolated query is outside C, and there is a lifted phase sequence with

$$x_n = r(s_n), \qquad a_n = s_n - s_{n-1}, \qquad 0 < a_{n+1} < a_n < a_* \quad (n \geq 0). \tag{4}$$

Put $d_n = x_n - x_{n-1}$ and $E_n = \|d_n\|^2$. Then

$$E_{n+1} + \|d_{n+1} - d_n\|^2 \leq E_n, \qquad d_n \longrightarrow 0, \tag{5}$$

and

$$\sum_{n=0}^{\infty} \|d_{n+1} - d_n\|^2 < \infty, \qquad \sum_{n=0}^{\infty} \|d_n\|^2 = \infty. \tag{6}$$

The exact asymptotic constants are

$$\sqrt{n}\, a_n \longrightarrow \sqrt{\frac{\ell}{2I}}, \qquad nE_n \longrightarrow \frac{\ell}{2I}, \tag{7}$$

$$\frac{s_n - s_0}{\sqrt{n}} \longrightarrow \sqrt{\frac{2\ell}{I}}, \qquad \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \|d_j\| \longrightarrow \sqrt{\frac{2\ell}{I}}. \tag{8}$$

Moreover, $(s_n - s_0) - \sum_{j=1}^{n} \|d_j\|$ has a finite limit, and

$$\omega(x_n) = \partial C, \qquad \omega((x_n, x_{n-1})) = \{(z, z) : z \in \partial C\}. \tag{9}$$

For every continuous $g : \partial C \to \mathbb{R}$,

$$\frac{1}{N} \sum_{n=0}^{N-1} g(x_n) \longrightarrow \frac{1}{\ell} \int_0^{\ell} g(r(s))\, \mathrm{d}s. \tag{10}$$

Thus vanishing increments coexist with indefinite travel and failure of point convergence. The limiting state set consists of fixed states on the boundary diagonal. Equation (10) gives the frequency of visits in the iteration index; it is a separate assertion from the limit-set statement.

The smallness threshold is uniform in the starting phase and orientation. The asymptotic statements concern each fixed $a_0 > 0$; they do not assert a uniform onset as $a_0 \downarrow 0$. Equal initial positions give a constant trajectory, and arbitrary interior initializations are outside the theorem. The energy E_n is the squared chord increment. If a fixed physical step $h > 0$ and unit mass are introduced, the chord-velocity kinetic convention is $E_n/(2h^2)$.

The map itself is classical. Paoli and Schatzman [1, Eq. (1.16)] consider the position scheme

$$U^{m+1} = -eU^{m-1} + (1+e)P_K\left(\frac{2U^m - (1-e)U^{m-1} + h^2 F^m}{1+e}\right). \tag{11}$$

For identity mass, $e = 0$ and $F^m = 0$, this is (1). Their Theorem 9.3 treats a vanishing-step limit on a finite physical-time interval. Here the initial pair and discrete update remain

fixed while the iteration index tends to infinity. The equation and theorem numbering just cited refer to the 1999 preprint.

The force-free position-based update of Müller, Heidelberger, Hennix and Ratcliff [2, Section 3] also gives (1) when its position solve is one exact metric projection and its separate damping and velocity-correction stages are omitted. Indeed, predicting $p = x + hv$, projecting p, and resetting $v = (p - x)/h$ eliminates the stored velocity to give the two-position recurrence. This identification does not say that every iterative constraint solver in position-based dynamics computes P_C. Artificial numerical damping in those methods is established background; see Dinev, Liu and Kavan [3, Section 2]. Macklin and coauthors [6] study substepping and its effect on damping in XPBD; their force-free chord-reset update likewise gives (1) when the position correction is an exact metric projection. Acary [4, Sections 3.2–3.3] distinguishes the position scheme from velocity-level impact schemes and notes that the $e = 0$ position method does not impose the discrete stepwise Newton impact law.

Our result is a fixed-step boundary-circulation theorem with an exact curvature average. It neither introduces the projection scheme nor claims a new occurrence of numerical damping. Its proof is self-contained. The source comparison is limited to inspected primary texts; older constrained-integration analyses and all second-order sweeping variants have not been exhaustively cleared. In particular, the full text of the geodesic-relaxation work [5] was not available for theorem-level comparison. No complete priority claim is made.

2 The actual projection and its phase loss

For a closed convex set, the characterization of metric projection is

$$q = P_C y \quad \Longleftrightarrow \quad q \in C, \qquad \langle y - q, z - q \rangle \le 0 \quad (z \in C). \tag{12}$$

Applying this inequality to two projections and adding proves nonexpansiveness. We use it first to locate the actual projection, before deriving a phase equation.

Lemma 1. *There are $a_* > 0$, $c > 0$ and $D > 0$, depending only on C, such that for all $s \in \mathbb{R}$ and $0 < a < a_*$,*

$$P_C(2r(s) - r(s - a)) = r(s + B(s, a)), \tag{13}$$

where B is periodic in s and

$$B(s, a) = a - \kappa(s)^2 a^3 + O_C(a^4). \tag{14}$$

The remainder is uniform in s. The query in (13) lies outside C, and

$$ca^3 \le a - B(s, a) \le Da^3, \qquad \frac{a}{2} \le B(s, a) < a. \tag{15}$$

The same threshold may be used for both orientations.

Proof. Let $T(s) = r'(s)$ and let $N(s)$ be the outward unit normal. Positive curvature and convexity imply that every supporting line touches ∂C at only one point: otherwise the segment joining two contact points would lie in the boundary. Hence, for $0 < a < \ell/8$,

$$\langle r(s-a) - r(s), N(s)\rangle < 0, \qquad \langle 2r(s) - r(s-a) - r(s), N(s)\rangle > 0. \tag{16}$$

Write $y = 2r(s) - r(s-a)$ and $q = P_C y$. The second inequality shows that $y \notin C$, so $q \in \partial C$. Since $P_C r(s) = r(s)$, nonexpansiveness gives

$$\|q - r(s)\| \le \|r(s) - r(s-a)\| \le a. \tag{17}$$

We justify a uniform local phase for this actual point. Choose $0 < \delta < \ell/8$ so small that $\langle T(s+u), T(s)\rangle \ge 1/2$ whenever $|u| \le 2\delta$. Integration implies

$$\|r(s+b) - r(s)\| \ge |b|/2 \qquad (|b| \le 2\delta). \tag{18}$$

Compactness and embeddedness give

$$m_\delta := \min_{\substack{0\le s\le \ell \\ \delta \le |b| \le \ell/2}} \|r(s+b) - r(s)\| > 0. \tag{19}$$

For $a < m_\delta$, equations (17)–(19) give a unique lift $q = r(s+b)$ with $|b| < \delta$ and $|b| \le 2a$. No sign has yet been assigned to b.

The squared distance to y has a minimum at this boundary point. Thus

$$G(s,a,b) := \langle 2r(s) - r(s-a) - r(s+b), T(s+b)\rangle = 0. \tag{20}$$

Uniform Taylor estimates yield

$$2r(s) - r(s-a) - r(s+a) = -a^2 r''(s) + O_C(a^4), \tag{21}$$

$$T(s+a) = T(s) + a r''(s) + O_C(a^2). \tag{22}$$

Since $\langle r''(s), T(s)\rangle = 0$, we obtain

$$G(s,a,a) = -\kappa(s)^2 a^3 + O_C(a^4). \tag{23}$$

For $|u| \le 2a$, direct differentiation gives

$$\partial_b G(s,a,u) = -1 + \langle y - r(s+u), r''(s+u)\rangle = -1 + O_C(a), \tag{24}$$

since $\|y - r(s+u)\| \le a + |u| \le 3a$. After shrinking a_*, this derivative is at most $-1/2$. The mean value theorem between the actual root b and the comparison value a proves

$$b - a = -\frac{G(s,a,a)}{\partial_b G(s,a,\xi)} = -\kappa(s)^2 a^3 + O_C(a^4). \tag{25}$$

Set $B(s,a)=b$. This proves the expansion for the actual metric projection rather than for a freely chosen stationary normal foot.

The positive minimum of κ^2 and its finite maximum allow fixed $c,D>0$ such that the first two inequalities in (15) hold throughout a sufficiently small interval. Further impose $Da_*^2\le 1/2$ to obtain its final bounds. All derivative and separation estimates are uniform in s modulo ℓ. Applying the argument in both orientations and taking the smaller threshold proves the last assertion. ∎

Starting from (3), define $a_{n+1}=B(s_n,a_n)$ and $s_{n+1}=s_n+a_{n+1}$. Lemma 1 applies inductively for every n. This proves exteriority of all queries, boundary persistence and (4). In particular the subsequent phase analysis describes the literal recurrence (1).

3 Dissipation and preliminary decay

The projection inequality gives an energy estimate before any averaging is used.

Proposition 3.1. For the orbit in Theorem 1,

$$E_{n+1}+\|d_{n+1}-d_n\|^2\le E_n\qquad (n\ge 0). \tag{26}$$

There are positive constants c,D depending only on C such that, with $t_n=a_n^{-2}$,

$$t_0+2cn\le t_n\le t_0+8Dn, \tag{27}$$

and

$$t_{n+1}-t_n=2\kappa(s_n)^2+O_C(a_n). \tag{28}$$

Consequently $a_n\asymp (n+1)^{-1/2}$, $s_n\to+\infty$, and $\sum_{n\ge0}a_n^3<\infty$. The comparison constants may depend on the fixed initial displacement.

Proof. In (12), take $y=2x_n-x_{n-1}$, $q=x_{n+1}$ and $z=x_n$. Since $y-q=d_n-d_{n+1}$ and $z-q=-d_{n+1}$, the resulting sign is

$$\langle d_n-d_{n+1},d_{n+1}\rangle\ge 0. \tag{29}$$

The identity

$$\|d_n\|^2-\|d_{n+1}\|^2-\|d_n-d_{n+1}\|^2=2\langle d_n-d_{n+1},d_{n+1}\rangle \tag{30}$$

proves (26). Summing also gives

$$\sum_{n=0}^{N}\|d_{n+1}-d_n\|^2\le E_0-E_{N+1}\le E_0. \tag{31}$$

Let $u_n=(a_n-a_{n+1})/a_n$. Lemma 1 gives

$$ca_n^2\le u_n\le Da_n^2\le\frac12,\qquad t_{n+1}-t_n=a_n^{-2}\big((1-u_n)^{-2}-1\big). \tag{32}$$

For $0 \le u \le 1/2$, elementary algebra yields

$$2u \le (1-u)^{-2} - 1 = \frac{2u - u^2}{(1-u)^2} \le 8u. \tag{33}$$

Hence $2c \le t_{n+1} - t_n \le 8D$, which sums to (27). This implies the stated comparison, the divergence of $\sum a_n$ and the convergence of $\sum a_n^3$. In particular $s_n - s_0 = \sum_{j=1}^{n} a_j \to \infty$.

For the sharper increment, equation (14) gives $u_n = \kappa(s_n)^2 a_n^2 + O_C(a_n^3)$. Substitute it into $(1-u)^{-2} = 1 + 2u + O(u^2)$ to obtain (28). ∎

The term $\kappa(s_n)^2$ in (28) is sampled at the actual changing phases. Its replacement by an arclength average requires an argument. Neither the decay estimate nor eventual visits to every boundary arc alone gives the iteration frequencies.

4 Sampling the boundary

We give a deterministic averaging statement adapted to the decreasing phase advances. It does not use a rotation number or a mixing hypothesis.

Lemma 2. *Let $h : \mathbb{R} \to \mathbb{R}$ be continuous, ℓ-periodic, and have zero mean over one period. For the phases of Theorem 1,*

$$\frac{1}{N} \sum_{j=0}^{N-1} h(s_j) \longrightarrow 0. \tag{34}$$

If h is Lipschitz, then

$$\sum_{j=0}^{N-1} h(s_j) = O(\sqrt{N}). \tag{35}$$

Proof. The primitive $H(s) = \int_0^s h(v)\,\mathrm{d}v$ is periodic and bounded. Put $q_j = a_{j+1}$ and $w_j = q_j^{-1}$. Since $s_{j+1} - s_j = q_j$, uniform continuity gives

$$w_j\big(H(s_{j+1}) - H(s_j)\big) = h(s_j) + e_j, \qquad |e_j| \le \omega_h(q_j), \tag{36}$$

where ω_h is a modulus of continuity. Exact summation by parts reads

$$\sum_{j=0}^{N-1} w_j\big(H(s_{j+1}) - H(s_j)\big) = w_{N-1}H(s_N) - w_0 H(s_0) - \sum_{j=1}^{N-1} (w_j - w_{j-1}) H(s_j). \tag{37}$$

The weights increase, because the advances decrease. Therefore the absolute value of (37) is at most

$$2\|H\|_\infty w_{N-1} = \frac{2\|H\|_\infty}{a_N} = O(\sqrt{N}), \tag{38}$$

by (27). Moreover, $q_j \to 0$, so the Cesàro mean of $\omega_h(q_j)$ tends to zero. Sum (36), divide by N, and use (38) to prove (34).

If h has Lipschitz constant L_h, integrating $|h(s_j+v)-h(s_j)| \le L_h v$ gives $|e_j| \le L_h q_j/2$. Equation (27) implies $\sum_{j=0}^{N-1} q_j = O(\sqrt{N})$. Together with (38), this proves (35). ∎

For continuous g on the boundary, apply (34) to

$$h(s) = g(r(s)) - \frac{1}{\ell}\int_0^\ell g(r(v))\,\mathrm{d}v. \tag{39}$$

This proves the occupation law (10). Since κ^2 is continuously differentiable and periodic, the quantitative assertion applies to its centered version. With $A = I/\ell$, it gives

$$\sum_{j=0}^{n-1} \kappa(s_j)^2 = An + O(\sqrt{n}). \tag{40}$$

In particular the relevant average is with respect to arclength even though individual steps lose more phase where curvature is larger.

5 Exact constants and the limit set

We now complete the proof of Theorem 1. Summing (28) and using (40) gives

$$t_n = t_0 + 2\sum_{j=0}^{n-1} \kappa(s_j)^2 + O\left(\sum_{j=0}^{n-1} a_j\right) = 2An + O(\sqrt{n}). \tag{41}$$

Thus, with $\gamma = (2A)^{-1/2} = \sqrt{\ell/(2I)}$,

$$\sqrt{n}\,a_n \longrightarrow \gamma. \tag{42}$$

Summing the comparison $a_j \sim \gamma j^{-1/2}$ yields

$$\frac{s_n - s_0}{\sqrt{n}} = \frac{1}{\sqrt{n}}\sum_{j=1}^{n} a_j \longrightarrow 2\gamma. \tag{43}$$

For example, for every $\varepsilon > 0$ the late terms lie between $(\gamma-\varepsilon)j^{-1/2}$ and $(\gamma+\varepsilon)j^{-1/2}$. The finite prefix vanishes after division by $\sqrt{n}$, and $n^{-1/2}\sum_{j=1}^{n} j^{-1/2} \to 2$.

To transfer these statements to chord increments, we keep the geometric error explicit.

Lemma 3. *Let $K = \max_s \kappa(s)$. For every boundary arc of length $a > 0$,*

$$0 \le a^2 - \|r(s) - r(s-a)\|^2 \le \frac{K^2 a^4}{12}, \tag{44}$$

$$0 \le a - \|r(s) - r(s-a)\| \le \frac{K^2 a^3}{12}. \tag{45}$$

Proof. Since $r(s) - r(s-a) = \int_{s-a}^{s} T(u)\,\mathrm{d}u$ and $\|T(u)\| = 1$, expansion of the square gives

$$a^2 - \|r(s) - r(s-a)\|^2 = \frac{1}{2}\int_{s-a}^{s}\int_{s-a}^{s} \|T(u) - T(v)\|^2\,\mathrm{d}u\,\mathrm{d}v. \tag{46}$$

The tangent is K-Lipschitz. The double integral of $(u-v)^2$ on a square of side a is $a^4/6$, proving (44). Divide by $a + \|r(s) - r(s-a)\| \geq a$ to obtain (45). ∎

Equations (42) and (44) imply

$$nE_n = na_n^2 + O(na_n^4) \longrightarrow \gamma^2 = \frac{\ell}{2I}. \tag{47}$$

Furthermore, by (45) and $\sum a_j^3 < \infty$,

$$0 \leq \sum_{j=1}^{n}\bigl(a_j - \|d_j\|\bigr) \leq \frac{K^2}{12}\sum_{j=1}^{\infty} a_j^3 < \infty. \tag{48}$$

The partial sums on the left increase, so they have a finite limit. Together with (43), this proves the second limit in (8). It also proves the stated bounded cumulative difference between arclength and chord length.

We have $\|d_n\| \leq a_n \to 0$, while (31) proves square summability of $d_{n+1} - d_n$. Since the constant in (47) is positive, comparison with the harmonic series proves $\sum E_n = \infty$. Thus both parts of (6) hold; divergence of the first-increment energy follows from the asymptotic law, rather than from the descent inequality alone.

It remains to determine the full limit sets. All positions lie on the compact boundary. Conversely, fix $z = r(t) \in \partial C$. For every sufficiently large integer k, let m_k be the first index such that $s_{m_k} \geq t + k\ell$. It exists because $s_n \to +\infty$. Minimality gives

$$0 \leq s_{m_k} - (t + k\ell) < a_{m_k} \longrightarrow 0. \tag{49}$$

The indices tend to infinity, so $x_{m_k} \to r(t) = z$. Hence $\omega(x_n) = \partial C$. Since $d_n \to 0$, every two-position state limit is diagonal; the same subsequences show that every (z, z) on the boundary diagonal occurs. This proves (9) and finishes the theorem.

6 A disk and the scope of the conclusion

For a disk of radius R, identify the plane with $\mathbb{C}$ and put its center at the origin. In coordinates adapted to the chosen direction, write $x_n = Re^{i\phi_n}$ and $\theta_n = a_n/R$. The actual query is

$$2x_n - x_{n-1} = Re^{i\phi_n}(2 - e^{-i\theta_n}), \qquad |2 - e^{-i\theta_n}|^2 = 5 - 4\cos\theta_n > 1. \tag{50}$$

Projection onto the filled disk is radial normalization, giving exactly

$$\theta_{n+1} = \arg(2 - e^{-i\theta_n}) = \arctan\left(\frac{\sin\theta_n}{2 - \cos\theta_n}\right). \tag{51}$$

The small positive branch is selected by the close-pair initialization. Its expansion is

$$\theta_{n+1} = \theta_n - \theta_n^3 + \frac{5}{4}\theta_n^5 + O(\theta_n^7). \tag{52}$$

Here $\ell = 2\pi R$ and $I = 2\pi/R$. Therefore the general result gives

$$n\theta_n^2 \longrightarrow \frac{1}{2}, \qquad nE_n = 4R^2 n \sin^2(\theta_n/2) \longrightarrow \frac{R^2}{2}. \tag{53}$$

This calculation checks both the local coefficient and the radius normalization. The general theorem requires averaging the spatially varying curvature; radial normalization alone does not provide that step.

The mechanism consists of a cubic phase loss, a decreasing advance of order $n^{-1/2}$, and a divergent cumulative phase. A bounded-primitive argument controls the actual sampling of the boundary and identifies the curvature average. These facts concern the exact position recurrence (1). They do not assign its chord velocity to the tangent space, identify it with a billiard collision map, or establish a discrete Newton impact law. Positive curvature and the prescribed boundary-pair neighborhood are retained throughout; no conclusion is asserted for flat boundary segments, zero initial displacement, or arbitrary initial states.

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